Atish Kumar Sharma

Github LinkedIn Kaggle Hugging Face atish.sharma7321@gmail.com +91-7321890564 Portfolio

PROFESSIONAL SUMMARY

AI/ML Engineer with 1 year of experience building production-grade **Generative AI** systems, including **AI agent and workflow automation platforms**, **RAG pipelines**, and **multi-agent orchestration**. M.Sc. in AI & ML from **IIIT Lucknow**, with a strong foundation in ML, DL, NLP, Computer Vision, and MLOps. Experienced in fine-tuning LLMs and building end-to-end AI applications with **Python, FastAPI, and Docker**.

EDUCATION

- **Indian Institute of Information Technology** — Lucknow, India
  Master of Science in AI and ML — Aug 2024 – May 2026
- **Patliputra University** — Patna, India
  B.Sc - Mathematics (Honours) — Aug 2019 – Sept 2022

EXPERIENCE

AI/ML Engineer – Qubittron (via Vedanga Softsol) *Feb 2026 – Present*

- Developing an **enterprise AI agent and workflow automation platform** with a visual node-based workflow builder, **RAG-powered Knowledge Bases** (hybrid search, reranking, tabular retrieval), and **multi-agent orchestration** with **Agent-to-Agent (A2A)** communication over JSON-RPC.
- Built a **database management and Workbench module** using **DuckDB**, enabling AI agents and workflows to query, combine, and analyze large **CSV/XLSX datasets** and multiple managed database sources.
- Implemented **role-based access control** across frontend and backend for workflows, agents, chats, and KB, shipping production features with **TypeScript, React, PostgreSQL, Drizzle ORM, Redis, and Docker**.

AI Intern – Edunet Foundation *Apr 2025 – Jul 2025*

- Built and deployed **LLM-based chatbots**, **RAG systems**, and **multi-agent AI applications**.
- Fine-tuned open-source LLMs using **LoRA/PEFT** and optimized prompts for production use.

PROJECTS

- **Post-Discharge Medical AI Assistant** [GitHub]
  Tech Stack: LangGraph, Groq API, ChromaDB, SQLite, FastAPI, Docker, Hugging Face Spaces
  - Built a **multi-agent RAG system** for post-discharge nephrology patient care using LangGraph.
  - Indexed a **1,547-page medical textbook** into **9,633 semantic chunks** and integrated **Groq LLM**.
  - Exposed inference via **FastAPI**, containerized with **Docker**, and deployed on **Hugging Face Spaces**.
- **Gemma-2B LoRA Fine-tuning (Quotes Generator)** [GitHub]
  Tech Stack: Python, Hugging Face, PyTorch, LoRA, Google Colab
  - Fine-tuned **Gemma-2B** using **LoRA** with **4-bit quantization** for memory-efficient training.
  - Performed supervised fine-tuning on a custom quotes dataset and deployed inference for text generation.
- **Agriculture AI Crop Ranking System using LSTM** [GitHub]
  Tech Stack: Python, LSTM, Scikit-learn, Pandas, NumPy, Matplotlib (Kaggle GPU)
  - Developed an AI-based crop ranking system using **7 soil and environmental features**.
  - Performed EDA, correlation analysis, and outlier handling using the **IQR method**.
  - Trained an **LSTM-based multi-class model** achieving **99.77% validation accuracy** 22 crop categories.

TECHNICAL SKILLS

- **Languages & Tools:** Python, C, SQL, GitHub, FastAPI, CI/CD Pipelines, HTML, CSS, JavaScript (Basic)
- **Libraries & Frameworks:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, OpenCV, React, Node.js
- **AI & Applied ML:** Machine Learning, Deep Learning, Transformers & LLMs, RAG Pipelines, Agentic AI, Prompt Engineering, LLM Fine-tuning (LoRA, QLoRA, PEFT), Model Evaluation, MLOps, Claude Code

ACHIEVEMENTS & CERTIFICATIONS

IIT JAM 2024: AIR 1531(Maths) — **Amazon ML Challenge**: Team AIR 503 — **Data Science** (Certificate)